- echo "this is demo deploy job"

8. once it is committed then we can see click that we can see the pipline ...

#2353268200

Experiment No. 03 Date 16-02-2026 Page No. 06

Name of Experiment

creating and pushing java file to github and configuring Jenkins

$ cd . .

$ cd Administrator

$ cd Desktop

$ cd Devops

→ go to VS code → run the Java code → same as Myfile [OR]

→ go to notepad → Paste code → save as Myfile in Desktop

→ go to cmd → net stop jenkins → net start jenkins

$ git clone https://github.com/thrisha26/prog5.git

$ git init

$ git add prog5.java

$ git config --global user.email "thrisha2620126@gmail.com"

$ git config --global user.name "thrisha-26"

$ git commit -m "this is my java file"

$ git push

deploy:
script:
- echo "this is demo deploy job"

8. once it is committed then we can see click that we can see the pipline

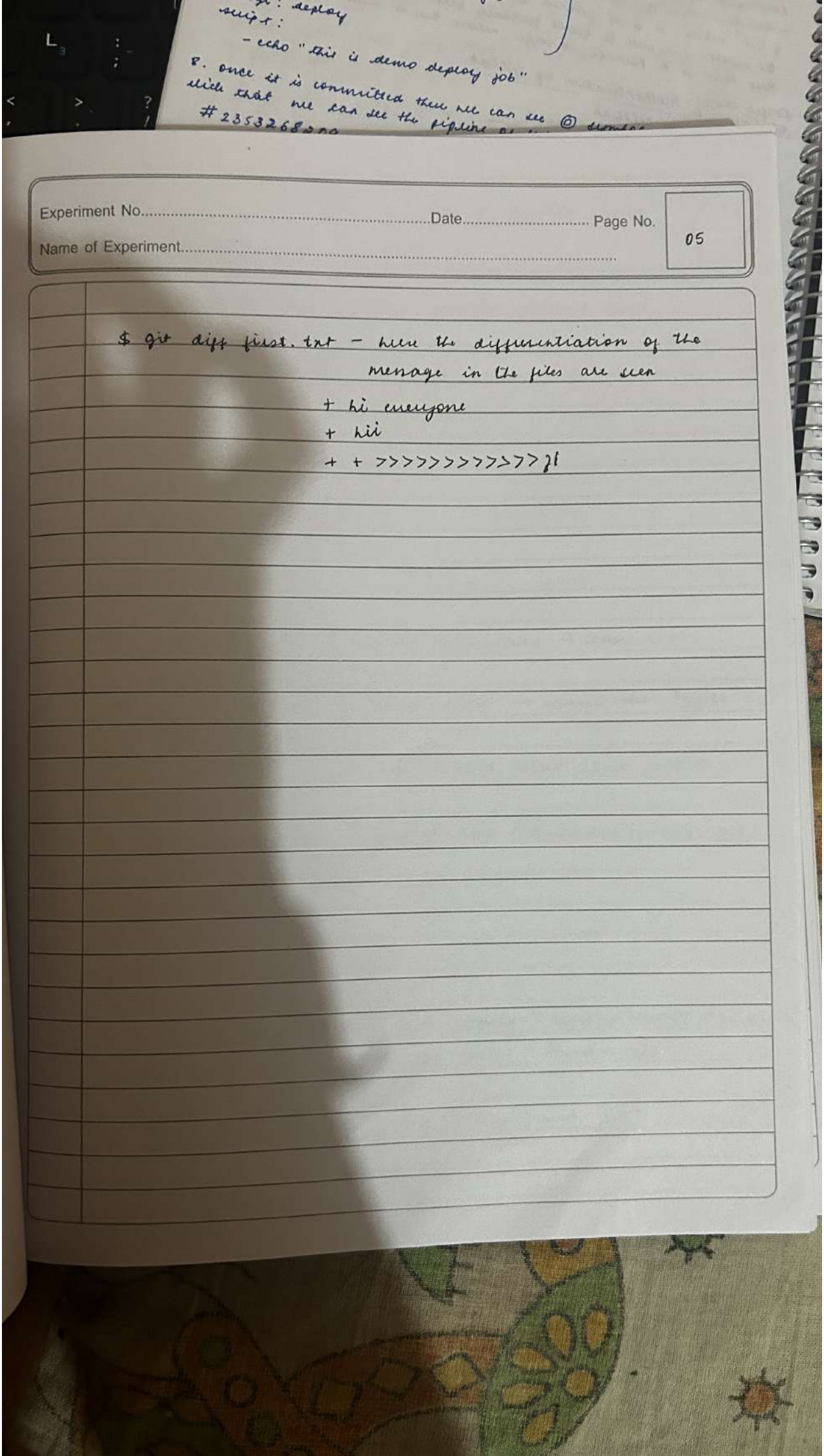
#2353268

Experiment No. ..... Date ..... Page No. 05

Name of Experiment .....

$ git diff first.txt - here the differentiation of the message in the files are seen

+ hi everyone
+ hii
+ + >>>>>>>>>>>>>>>>



- echo "this is demo deploy job"

8. once it is committed then we can see click that we can see the pipline or ... ⓪ number #2353268200

Experiment No.....Date..... Page No. 04

Name of Experiment.....

$ git commit -m "file 2" - message is saved to local repository

$ git checkout main - this is specified if master give master if main give main

file1.txt | 1 +
file2.txt | 1 +

$ git merge f1 - combine changes from 1 branch into another

$ nano first.txt - message: "Hi everyone"

$ git add first.txt - copy of "first.txt"

$ git commit -m "first" - message is saved to local repository

$ git checkout f1 - switched to f1

$ nano first.txt - give different message "hi"

$ git add first.txt - copy of "first.txt" to working directory

$ git commit -m "first-f1" - give different name

$ git checkout main => switched to branch 'main'

$ git merge f1 - CONFLICT message is poped
Automatic merge failed

- echo "this is demo deploy job"

8. once it is committed then we can see ⓪ symbol
click that we can see the pipline or ...
#2353268200

Experiment No. 02 Date 31-01-26 Page No. 03

Name of Experiment

2. Demonstrate to create a project in remote repository then apply fork, merge, diff, merge conflict, branch and pull request concepts on repository using github

1. go to github
2. search "Add two numbers"
3. select fork
4. then rename repository name → 'Add' → create fork
5. then click on code → copy url
6. go to git bash

```
$ cd ..
$ cd student-A
$ cd Desktop
$ cd Devops
$ git clone https://github.com/thisha-26/Add.git
$ cd Add
$ git init
$ git branch f1
$ git checkout f1      - switched to branch f
$ nano file1.txt       - message : "Hello" ctrl S + ctrl X
$ git add file1.txt -
                  copy of 'file1.txt' in working directory
$ git commit -m "file1" - message is saved to
                  local repository

$ nano file2.txt  - message : "Hello" ctrl S + ctrl X

$ git add file2.txt  - copy of 'file2.txt' in working
                  directory
```

"demo test job"
stage: deploy
script:
- echo "this is demo deploy job"

8. once it is committed then we can see click that we can see the pipline or u. ⓪ symbol
#2353268000

Experiment No.....Date..... Page No. 02
Name of Experiment.....

$ git add first.txt
if there are more files to be selected
$ git add .
$ git commit - m "first file"
$ git status
$ git push - a message is pop used → click sign in it will show the 404 error page

9. go to github → click program-1 → you can see first.txt → click on it → we can see message

10. in that can see edit option → put message like "hello everyone" then click on commit changes

11. go to gitbash
$ git pull
$ nano first.txt → a file will open
123456
good afternoon
hello everyone
click ctrl S + ctrl X

$ git reset -- hard
rest the messages

$ nano first.txt → we can check & see old message

$ git log

- echo "this is demo test job"
deploy job:
stage: deploy
script:
- echo "this is demo deploy job"

8. once it is committed then we can see ⓪ symbol
click that we can see the pipline of it.
#2353268209

Experiment No. 01 Date 27-01-26 Page No. 01

Name of Experiment

1. Demonstrate and create project in local and remote repository using GitBash and Github and apply init, status, log, add, commit, push, config, clone and reset commands on repository

1. create new folder 'Devops'
2. go inside and copy the address
   C: \users \student-A \Desktop \Devops
3. open git bash : Syntax
   $ cd . .
   $cd Student-A
   $ cd Desktop
   $ cd Devops

4. open github → new → name : Program-1 → create repository

5. copy the https://github.com/thisha-26/program-10.git ✓ copy
6. paste the https with
   $ git clone https://github
7. $ git init
   $ cd program-1
   $ nano first.txt
   - A file is opened → there type
   good afternoon
   then click ctrl S + ctrl X
8. $ git config --global user.email thisha@gmail.com
   $ git config --global user.name thisha-26

```
script:
  - echo "this is demo test job"
deploy job:
  stage: deploy
  script:
    - echo "this is demo deploy job"
```

8. once it is committed then we can see with that we can see the pipeline ...

#2353268200

Experiment No. ..... Date ..... Page No. 01

Name of Experiment .....

```
Package example1.example1;
import java.io.FileReader;
import java.io.IOException;
import java.text.ParseException;

import org.json.simple.JSONObject;
import org.json.simple.parser.JSONParser;

public class Example1 {
    public static void main (String[] args) throws IOException,
                        ParseException, org.json.simple.parser.ParseException {
  JSONParser jsonparser = new JSONParser();
  FileReader filereader = new FileReader(".\\jsonfile\\emp.json");
  Object obj = jsonparser.parse(filereader);
  JSONObject empobj = (JSONObject) obj;
  String fname = (String) empobj.get("firstname");
  String lname = (String) empobj.get("Lastname");
  System.out.println("Firstname :" + fname);
  System.out.println("Lastname :" + lname);
 }
}
```

→ Run

```
  - echo "this is demo build job"
test job:
  stage: test
  script:
    - echo "this is demo test job"
deploy job:
  stage: deploy
  script:
    - echo "this is demo deploy job"
```

commit message

commit

8. once it is committed then we can see click that we can see the pipeline ... demo

#2353268200

Experiment No. 04 Date 03-03-2026 Page No. 08

Name of Experiment

create a maven project with all dependencies required for application in CI/CD pipeline

1. Go to file → New → Maven project → next → In filter type → org.apache.maven. go to → artifact id → maven-archetype-quickstart - 1.5 → it should be 1.5 if not restart the eclipse
2. there give groupid : example1 & artifact id : example1 → next → console will open → type Y we can see BUILD SUCCESS and folder in corner
3. Go to → example1 → pom.xml → go to 33 line
4. Go to chrome → Maven Repository → in search type → simple json → Select 1st link → Go to lastest versions → select 1.1.1 → scroll down → copy the code inside dependency → paste it to 33 line → save
5. Select example1 → right click → new → folder → name it as jsonfile
   select example1 → right click → new → file → name it as emp.json

6. type under emp.json

```
{
    "firstname" : "Thrisha",
    "lastname" : "yonda"
}
```

7. select src / test / java → select → new → class → name → Example1 → finish
8. There write code.

(13). A file is opened & then type anything
good afternoon

then click ctrl S → ctrl X

(14). we can see the file in folder then type

(15). git config --global user.email thrisha262006@gmail.com

(16) git config --global user.name thrisha-26

(17). git add first.txt → here we have only 1 file that is first.txt it will be added

if there are many files then

git add . → all the files which is recording categories means current recording will be added

(18). git commit -m "first file"

(19). git status } use it before commit

(20) git push → a message is pop uped → click sign it will show the error message

(21) go to github → click program-1 → u can see first.txt click on it u can see message

(22). in that u can see edit option → put message like "hello everyone" then click on commit changes

(23). go to git bash

(24) git pull → if u go & see the local repository "we can see edited message there"

(25). nano first.txt → a file is opened → type numbers then

123456
good afternoon
hello everyone

click ctrl S & ctrl X

(26) git reset --hard → it will reset into back

(27). nano first.txt → a file is opened → there u can see the old message itself

(28) git log → it will entry all the details about the user

if you want to create another file then go from the step 10

git init

cd program-1

nano second.txt

*** AQ

```
resource " azurerm - resource - group " " my rg " {
  name = " MyAppResource group "
  location = " west Europe "
  tags = {
    environment = " Bookdemo "
  }
}
```

till here everything will be there

we have to change here

*** Another example that will allow you to install & restart nginx on a server using Ansible

```
- hosts : all
  tasks :
  - name : install and check nginx latest version
    apt : name : nginx state : latest
  - name : start nginx
    service
      name : nginx
      state : start → / stopped
```

till here it will be there

here we specify & start : start → / stopped we can make it to start / stop

27 Jan 2026 | Tuesday

LAB: Demonstrate & create a project in local repository and remote repository using git bash and github and apply tag init, status, log, add, commit, Push, conflict, clone and reset commands on repositories

1. create new folder 'Devops'
2. go inside and copy the address.
   C:\Users\student-A\Desktop\Devops
3. open git bash
   type cd ..
4. cd Student-A
5. cd Desktop
6. cd Devops
7. open github → new → name : Program-1 → create repository
8. copy the https://github.com/thrisha-26/Program-1.git → copy
9. past the https with
   git clone https://github
   duplicate folder will be established
10. git init
11. cd Program-1
12. nano first.txt

* cloning the repository : $ git clone url
1. creates the new directory in the name of
2. creates the local repository with its initialization & configuration
3. downloads the remote clone

* updating the code : $ git init
$ git add first.txt
$ git commit -m 'file'
$ git pull

* Retrieve the updates & archiving the remote repository :
$ git pull origin Master / main

→ consider a creates a 'myfile' he wants to push repo, name it as
" "

16 feb 2026
Monday

MODULE : 3

* implementation & installation of Jenkins
$ cd ..
$ cd student - A
$ cd desktop
$ cd Devops

→ create new repository → copy link
$ git clone url .....
→ go to VS code → run the java code → same as : myfile in desktop
~~in prog5~~
$ cd programs
$ ~~cd~~ git add ~~prog5.java~~ myfile.java
$ git commit -m " this is my java prog "
$ git push
→ he can go ∞ and see in git "The program is created here"

Jenkins → new item → enter an item name.

1. name - program5
2. select freestyle project → OK
3. configure → source code manag → Git
4. go to git copy the url and paste it
5. Branch to build → */main (edit it to main)
6. Build steps → execute windows batch command
   type : javac myfile.java
7. Save
8. corner → Build Now → here we can green at the bottom that it is right if not, red → wrong
9. #1 → select console output → there u can see SUCCESS

* If we do in Jenkin then there is no need of configuring again and again we can trigger it for execution for that reason we need webhook

Webhook

1. gitrep → settings
2. webhook → add webhook → payload URL (address of Jenkins : localhost:18080)
3. download ngrok → zip file is downloaded → inside the ngrok → type ngrok http 8080
4. copy the url then paste it to payload URL +/ url /program5/git → in build trigger ☑ github web hook trigger → save
6. go to git hub repository → program panel → configure something → commit changes
7. we can go & see in Jenkins → where we can see the green #2

Main game 2 sheets → read Man's code in last

consider so & so person wants to create a firstfile.java ~~firstfile.java print~~ u'll write & configure & execute selected git URL, what is the command for executing of Java console, write completely

installation & implementation of Jenkins

$ cd ..

$ cd Administrator go to notepad → paste code same as Myfile in desktop

$ cd Desktop or VS code → run the java code

$ cd Devops → go to → save as Myfile → go cmd → net stop jenkins net start jenkins

$ git clone https://github.com/thrisha-26/progs.git

$ git init

$ git add progs.java

$ git config --global user.email "thrisha262006@gmail.com"

$ git config --global user.name "thrisha-26"

$ git commit -m "this is my java file"

$ git push

Jenkins → download → next → password

Jenkins

1. $ cd ..

$ cd student

$ cd Desktop

$ cd Devops

$ git clone url → go to VS code & run code save as myfile.java

$ git init

$ cd prog 5

$ git add myfile.java

$ git commit - m "this is my java prog"

$ git push

we can go and see "the program is created here" in git

LAB 1. File → maven project → next → file type → org.apache.maven → go for artifact id → select maven-archetype-quickstart - 1.5 → it should have 1.5 itself

LAB: 1. Git Bash

2. will create repository in GitHub → copy repository link

3. clone it same file as in desktop → Devops → prog5

4. go to VS code → Myfile.java

5. Git Bash → add & push

6. Jenkins → all steps

1. make use of git commands to create a file in local repository with file name "sample.txt" and to push the same into remote repository → prog 1 → push it to the remote repo with help of commands

2. use git commands to create a java program in local repository, push it to remote github repository & configure a jenkins CI job to execute the task → prog 3

3. consider a developer creating a repository named "sample" with a text file "demo.txt" & setting up a webhook to alert jenkins whenever changes are pushed → webhook

4. Write the steps involved in creating project in gitlab CI pipeline along with yml file → gitlab

```
Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git add first.txt
warning: in the working copy of 'first.txt', LF will be replaced by CRLF the next time Git touches it

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git commit -m "firstfile"
[f1 2e5a092] firstfile
 1 file changed, 1 insertion(+)
 create mode 100644 first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git merge f1
Auto-merging first.txt
CONFLICT (add/add): Merge conflict in first.txt
Automatic merge failed; fix conflicts and then commit the result.

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main|MERGING)
$ git diff first.txt
diff --cc first.txt
index fbcdbe1,530ae8d..0000000
--- a/first.txt
+++ b/first.txt
@@@ -1,1 -1,1 +1,5 @@@
++<<<<<<< HEAD
 +Hai Everyone
++=======
+ hai good morning
++>>>>>>> f1

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main|MERGING)
```

MINGW64:/c/Users/Student-A/Desktop/DevOps/Add

```
 create mode 100644 file1.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ nano file2.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git add file2.txt
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git commit -m "file2"
[f1 5e9bcd1] file2
 1 file changed, 1 insertion(+)
 create mode 100644 file2.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git merge f1
Updating e109400..5e9bcd1
Fast-forward
 file1.txt | 1 +
 file2.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 file1.txt
 create mode 100644 file2.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ nano first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git add first.txt
warning: in the working copy of 'first.txt', LF will be replaced by CRLF the next time Git touches it

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git commit -m "first"
[main 2e02d03] first
 1 file changed, 1 insertion(+)
 create mode 100644 first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git checkout f1
Switched to branch 'f1'

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ nano first.txt
```

MINGW64:/c/Users/Student-A/Desktop/DevOps/Add

```
Student-A@M308-004 MINGW64 ~
$ cd ..

Student-A@M308-004 MINGW64 /c/Users
$ cd Student-A

Student-A@M308-004 MINGW64 ~
$ cd Desktop

Student-A@M308-004 MINGW64 ~/Desktop
$ cd DevOps

Student-A@M308-004 MINGW64 ~/Desktop/DevOps (master)
$ git clone https://github.com/Sinchana765/Add.git
Cloning into 'Add'...
remote: Enumerating objects: 55, done.
remote: Counting objects: 100% (55/55), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 55 (delta 20), reused 37 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (55/55), 40.82 KiB | 10.21 MiB/s, done.
Resolving deltas: 100% (20/20), done.

Student-A@M308-004 MINGW64 ~/Desktop/DevOps (master)
$ cd Add

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git init
Reinitialized existing Git repository in C:/Users/Student-A/Desktop/DevOps/Add/.git/

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git branch f1

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (main)
$ git checkout f1
Switched to branch 'f1'

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ nano file1.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git commit -m "file1"
[f1 d3b5e0f] file1
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ nano file2.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/Add (f1)
$ git add file2.txt
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it
```

```
Student-A@M308-004 MINGW64 ~
$ cd ..

Student-A@M308-004 MINGW64 /c/Users
$ cd Student-A

Student-A@M308-004 MINGW64 ~
$ cd Desktop

Student-A@M308-004 MINGW64 ~/Desktop
$ cd DevOps

Student-A@M308-004 MINGW64 ~/Desktop/DevOps
$ git clone https://github.com/Sinchana765/prog1.git
Cloning into 'prog1'...
warning: You appear to have cloned an empty repository.

Student-A@M308-004 MINGW64 ~/Desktop/DevOps
$ git init
Initialized empty Git repository in C:/Users/Student-A/Desktop/DevOps/.git/

Student-A@M308-004 MINGW64 ~/Desktop/DevOps (master)
$ cd prog1

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ nano first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git config --global user.email vvce23cseaiml0114@vvce.ac.in

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git config --global user.name Sinchana765

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git add first.txt
warning: in the working copy of 'first.txt', LF will be replaced by CRLF the next time Git touches it

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git commit -m "first file"
[main (root-commit) 96a743f] first file
 1 file changed, 1 insertion(+)
 create mode 100644 first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git status
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

nothing to commit, working tree clean
```

```
Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git push
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 232 bytes | 116.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sinchana765/prog1.git
 * [new branch]      main -> main

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 914 bytes | 70.00 KiB/s, done.
From https://github.com/Sinchana765/prog1
   96a743f..97c8f6b  main       -> origin/main
Updating 96a743f..97c8f6b
Fast-forward
 first.txt | 1 +
 1 file changed, 1 insertion(+)

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ nano first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git reset --hard
HEAD is now at 97c8f6b Update first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ nano first.txt

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$ git log
commit 97c8f6bb6a97dfc01dddfb35fecdca14b2837c6d (HEAD -> main, origin/main, origin/HEAD)
Author: Sinchana765 <vvce23cseaiml0114@vvce.ac.in>
Date:   Tue Jan 27 16:10:08 2026 +0530

    Update first.txt

commit 96a743f1b61f06680921cbb6e9a516e6d8c97744
Author: Sinchana765 <vvce23cseaiml0114@vvce.ac.in>
Date:   Tue Jan 27 15:57:23 2026 +0530

    first file

Student-A@M308-004 MINGW64 ~/Desktop/DevOps/prog1 (main)
$
```